higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

15 August 2025 (X-paper)
09:00–12:00

Drawing instruments may be used.
Calculators may NOT be used.

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 How many flip-flops are needed for a module 7 counter?

A 2
B 4
C 3
D 5

1.1.2 Which adder is used to add three variables A, B and C?

A Decoder
B Full
C Half
D Shift

1.1.3 What is simplest form of the Boolean expression $(A+B)(A+C)$?

A 1
B A
C AB
D A + BC

1.1.4 How many numbers are in the octal number system?

A 15
B 7
C 8
D 10

1.1.5 The error detection code is an ….

A 2421 BCD code
B Hamming code
C 5421 BCD code
D Grey code

(5 × 1) (5)

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1.2 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 A truth table illustrates how the input level of a gate responds to all the possible output level combinations.

1.2.2 As a rule, CMOS has the lowest power consumption of all IC families.

1.2.3 An inverter output is the complement of its input.

1.2.4 The LSB is always to the right of the MSB.

1.2.5 The falling edge of a signal is the transition from a HIGH to a LOW.

(5 × 1) (5)
[10]

QUESTION 2

2.1 Do the following calculations in the binary number system. Show ALL conversions and calculations clearly.

2.1.1 $14,5_8$ - $7,B_{16}$

Make use of one's complement and convert the final answer to decimal number. (6)

2.1.2 $2C,5_{16} \div 6,25_{10}$

Convert the final answer to an octal number. (6)

2.1.3 $12,125_{10} \times 5,4_8$

Convert the final answer to a hexadecimal number (6)

2.2 Convert the following code as indicated:

2.2.1 $11110010,101_{grey}$ to an octal number.

2.2.2 11001011100_{xs3} to a decimal number.

(2 × 3) (6)
[24]

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QUESTION 3

3.1 Write down the simplest form of the following Boolean expressions.

3.1.1 $AB + A\overline{B}$ (1)

3.1.2 $X(\overline{X} + Y)$ (1)

3.1.3 $A + A\overline{B} + ABC$ (1)

3.2 Simplify the following Boolean expression:

3.2.1 $A(A+B)(\overline{A}+\overline{B}+C) = F$ (6)

3.2.2 $Z(X+\overline{Y}) + X(Y+\overline{Z}) + \overline{X}Y = F$ (6)

3.3 Draw the circuit diagram of the Boolean expression below. Do NOT simplify the expression:

3.3.1 $\overline{\overline{(BC+CD)}\ C} + \overline{B+D} = F$ (6)

3.3.2 $\overline{\overline{X\overline{Y} \oplus Y\overline{Z}} + X\overline{Y}Z} = F$ (5)

[26]

QUESTION 4

4.1 Draw a neat, labelled circuit diagram of a 4-bit JK right shift register. Clearly indicate how to shift 1 through a 4-bit right shift by make use of the truth table. (10)

4.2 Draw a neat, labelled circuit diagram of a serial binary adder. (7)

4.3 Draw a neat, labelled circuit symbol for a 4 × 1 multiplexer. (3)

[20]

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QUESTION 5

5.1 Draw a neat, labelled circuit diagram of 4-bit synchronous up-counter. (8)

5.2 Give TWO advantages of a CMOS circuit. (2)

5.3 Define the following terms:

5.3.1 Noise immunity (2)

5.3.2 Logic swing (2)

5.4 Draw a block diagram of a successive approximation A/D converter (6)
[20]

TOTAL: 100

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